CSCE 222

Homework 5

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Contents

0.1 12 - Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$ if

P(n) = $a_n = -3a_n - 1 + 4a_{n-2}$ for all n≥2

0.1.1 a - $a_n = 0$

Basis step: $P(2) = a_2 = 0 = -3(0) + 4(0) = 0$ which holds

Inductive step: $p(k) \to p(k+1)$

Assume p(k) holds for k ≥ 2, $a_k = 0, a_{k-1} = 0$

LHS: $a_{k+1} = 0$

RHS: $-3a_k + 4a_{k-1}$

$-3(0) + 4(0) = 0$

LHS = RHS therefore $p(k) \to p(k+1)$ holds for all n ≥ 2 by induction.

0.1.2 b - $a_n = 1$

Basis step: $P(2) = a_2 = 1 = -3(1) + 4(1) = 1$ which holds

Inductive step: $p(k) \to p(k+1)$

Assume p(k) holds for k $\geq$ 2, $a_k = 1, a_{k-1} = 1$

LHS: $a_{k+1} = 1$

RHS: $-3a_k + 4a_{k-1}$

$-3(1) + 4(1) = 1$

LHS = RHS therefore $p(k) \to p(k+1)$ holds for all n $\geq$ 2 by induction.

0.1.3 c - $a_n = (-4)^n$

Basis step: $P(2) = a_2 = (-4)^2 = 16 = -3((-4)^1) + 4((-4)^0) = 12 + 4 = 16$ which holds

Inductive step: $p(k) \to p(k+1)$

Assume p(k) holds for k $\geq$ 2, $a_k = -4^k, a_{k-1} = -4^{k-1}$

LHS: $a_{k+1} = (-4)^{k+1}$

RHS: $-3a_k + 4a_{k-1}$

$-3((-4)^k) + 4((-4)^{k-1})$

$-3((-4)^k) + 4((-4)^k(-4)^{-1})$

$-3((-4)^k) + ((-4)^k(-1))$

$-4(-(4)^k)$

$(-4)^1((-4)^k)$

$((-4)^{k+1})$

LHS = RHS therefore $p(k) \to p(k+1)$ holds for all n $\geq$ 2 by induction.

0.1.4 d - $a_n = 2(-4)^n + 3$

Basis step: $P(2) = a_2 = 2(-4)^2 + 3 = 35 = -3(2(-4)^1 + 3) + 4(2(-4)^0 + 3) = 15 + 20 = 35$ which holds

Inductive step: $p(k) \to p(k+1)$

Assume p(k) holds for k $\geq$ 2, $a_k = 1, a_{k-1} = 1$

LHS: $2(-4)^{k+1} + 3$

RHS: $-3a_k + 4a_{k-1}$

$-3(2(-4)^k + 3) + 4(2(-4)^{k-1} + 3)$

$(-6(-4)^k + -9) + (8(-4)^k(-4)^{-1} + 12)$

$(-6(-4)^k + -9) + ((-4)^k(-2) + 12)$

$(-6(-4)^k) + (-2(-4)^k) + 3$

$(-8(-4)^k) + 3$

$((2(-4)^1)(-4)^k) + 3$

$2(-4)^{k+1} + 3$

LHS = RHS therefore $p(k) \rightarrow p(k+1)$ holds for all n $\geq$ 2 by induction.

0.2 Question 14 - Find the recurrence relation satisfied by the sequence $\{a_n\}$

0.2.1 a - $a_n = 0$

$a_n = 0(a_{n-1})$ for all n $\geq$ 1, where $a_0 = 0$

0.2.2 b - $a_n = 1$

$a_n = (a_{n-1})$ for all n $\geq$ 1, where $a_0 = 1$

0.2.3 c - $a_n = 2^n$

$a_n = 2(a_{n-1})$ for all n $\geq$ 1, where $a_0 = 1$

0.2.4 d - $a_n = 4^n$

$a_n = 4(a_{n-1})$ for all n $\geq$ 1, where $a_0 = 1$

0.2.5 e - $a_n = n4^n$

a_n = 8(a_{n-1}) -16(a_{n-2}) for all n $\geq$ 2, where a_0 = 0, a_1 = 4

0.2.6 f - $a_n = 2(4^n) + 3n4^n$

a_n = 8(a_{n-1}) -16(a_{n-2}) for all n $\geq$ 2, where a_0 = 0, a_1 = 4

0.2.7 g - $a_n = (-4)^n$

a_n = -4(a_{n-1}) for all n $\geq$ 1, where a_0 = 1

0.2.8 h - $a_n = n^2 4^n$

a_n = 12(a_{n-1}) -48(a_{n-2}) + 64(a_{n-3}) for all n $\geq$ 3, where a_0 = 0,a_1 = 4, a_2 = 64

0.3 Question 22a - An employee joined a company in 2023 with a starting salary of $50,000. Every year this employee receives a raise of $1000 plus 5% of the salary of the previous year.

0.3.1 a - Set up a recurrence relation for the salary of this employee n years after 2023.

a_0 = 50,000

a_n = a_{n-1} + 1,000 + 0.05a_{n-1}

a_n = 1.05a_{n-1} + 1,000 for all n$\geq$1, where a_0 = 50,000

0.4 Question 22b - What will the salary of this employee be in 2031?

n = 2031 - 2023 = 8

a_n = 1.05a_{n-1} + 1,000 for all n≥1, where a_0 = 50,000

a_1 = 1.05a_0 + 1,000 = 53,500 in 2024

a_2 = 1.05a_1 + 1,000 = 57,175 in 2025

a_3 = 1.05a_2 + 1,000 = 61,033.75 in 2026

a_4 = 1.05a_3 + 1,000 = 65,085.44 in 2027

a_5 = 1.05a_4 + 1,000 = 69,339.71 in 2028

a_6 = 1.05a_5 + 1,000 = 73,806.70 in 2029

a_7 = 1.05a_6 + 1,000 = 78,497.04 in 2030

a_8 = 1.05a_7 + 1,000 = 83,421.89 in 2031

Thus, the salary in 2031 would be \$83,421.89

0.5 Question 32 - Find the value of each of these sums.

0.5.1 a-$\sum_{j=0}^{8} 1 + (-1)^j$

$\sum_{j=0}^{8} 1 + (-1)^j$ = $\sum_{j=0}^{8} 1$ + $\sum_{j=0}^{8} (-1)^j$
$\sum_{j=0}^{8} 1 + (-1)^j$ = 9(1) + 1 + 4(-1 + 1) = 10

0.5.2 b-$\sum_{j=0}^{8}(3^j - 2^j)$

$\sum_{j=0}^{8}(3^j - 2^j)$ = $\sum_{j=0}^{8}(3^j)$ - $\sum_{j=0}^{8}(2^j)$
Using the geometric series formula:
$\sum_{j=0}^{8}(3^j - 2^j)$ = $(\frac{3^0(1-3^9)}{1-3})$ - $(\frac{2^0(1-(2)^9)}{1-2})$
$\sum_{j=0}^{8}(3^j - 2^j)$ = (9841 - 511) = 9330

0.5.3 c-$\sum_{j=0}^{8}(2 \cdot 3^j + 3 \cdot 2^j)$

Using the geometric series formula:
$\sum_{j=0}^{8}(2 \cdot 3^j + 3 \cdot 2^j)$ = $2(\frac{3^0(1-3^9)}{1-3})$ + $3(\frac{2^0(1-(2)^9)}{1-2})$

using part b,

$\sum_{j=0}^{8}(2 \cdot 3^j + 3 \cdot 2^j) = 2(9841) + 3(511) = 21215$

0.5.4 d-$\sum_{j=0}^{8}(2^{j+1} - 2^j)$

$\sum_{j=0}^{8}(2^{j+1} - 2^j) = \sum_{j=0}^{8}(2^j 2 - 2^j)$

$\sum_{j=0}^{8}(2^{j+1} - 2^j) = (2\sum_{j=0}^{8} 2^j) - (\sum_{j=0}^{8}(2^j)) = (\sum_{j=0}^{8}(2^j))$

Using the geometric series formula:

$\sum_{j=0}^{8}(2^{j+1} - 2^j) = \frac{1-2^9}{1-2} = 511$

1 Section 5.1

1.1 Question 8 - Prove that $2 - 2 \cdot 7 + 2 \cdot 72 - \ldots + 2(-7)^n = (1 - (-7)^{n+1})/4$ whenever n is a nonnegative integer.

P(n) = $2 - 2 \cdot 7 + 2 \cdot 72 - \ldots + 2(-7)^n = (1 - (-7)^{n+1})/4$ for all n≥0

Basis step: $P(0) = a_0 = 2(-7)^0 = 2 = (1 - (-7)^{0+1})/4 = 2$ which holds

Inductive step: $p(k) \to p(k+1)$

Assume p(k) holds for k ≥ 0, $2 - 2 \cdot 7 + 2 \cdot 72 - \ldots + 2(-7)^k = (1 - (-7)^{k+1})/4$

RHS: $(1 - (-7)^{(k+1)+1})/4$

LHS: $(1 - (-7)^{k+1})/4 + a_{k+1}$

$(1 - (-7)^{k+1})/4 + 2(-7)^{k+1}$

$(1 - (-7)^{k+1})/4 + (8(-7)^{k+1})/4$

$(1 + (8 - 1)(-7)^{k+1})/4$

$(1 + 7(-7)^{k+1})/4$

$(1 - -7(-7)^{k+1})/4$

$(1 - (-7)^{(k+1)+1})/4$

LHS = RHS therefore $p(k) \to p(k+1)$ holds for all n ≥ 0 by induction.

1.2 Question 12 - Prove $\sum_{j=0}^{n}(-\frac{1}{2})^j = \frac{2^{n+1}+(-1)^n}{3\cdot 2^n}$ for n $\geq$ 0

P(n) = $\sum_{j=0}^{n}(-\frac{1}{2})^j = \frac{2^{n+1}+(-1)^n}{3\cdot 2^n}$ for all n $\geq$ 0

Basis step: $P(0) = a_0 = \sum_{j=0}^{0}(-\frac{1}{2})^0 = 1 = \frac{2^{0+1}+(-1)^0}{3\cdot 2^0} = \frac{3}{3} = 1$ which holds

Inductive step: $p(k) \to p(k+1)$

Assume p(k) holds for k $\geq$ 0, $\sum_{j=0}^{k}(-\frac{1}{2})^j = \frac{2^{k+1}+(-1)^k}{3\cdot 2^k}$

RHS: $\frac{2^{(k+1)+1}+(-1)^{k+1}}{3\cdot 2^{k+1}}$

LHS: $\sum_{j=0}^{k}(-\frac{1}{2})^j + (-\frac{1}{2})^{k+1}$

$\frac{2^{k+1}+(-1)^k}{3\cdot 2^k} + (-\frac{1}{2})^{k+1}$

$\frac{2(2^{k+1}+(-1)^k)}{3\cdot 2^{k+1}} + (\frac{3(-1)^{k+1}}{3\cdot 2^{k+1}})$

$\frac{(2(2^{k+1})-(-1)^k)}{3\cdot 2^{k+1}}$

$\frac{(2^{(k+1)+1}+(-1)^{k+1})}{3\cdot 2^{k+1}}$

LHS = RHS therefore $p(k) \to p(k+1)$ holds for all n $\geq$ 0 by induction.

1.3 Question 14 - Prove that for every positive integer n, $\sum_{k=1}^{n}(k2^k) = (n-1)2^{n+1} + 2$

P(n) = $\sum_{k=1}^{n}(k2^k) = (n-1)2^{n+1} + 2$ for all n $\geq$ 1

Basis step: $P(1) = a_1 = \sum_{k=1}^{n}(k2^k) = 2 = (n-1)2^{n+1} + 2 = 0 + 2$ which holds

Inductive step: $p(k) \to p(k+1)$

Assume p(j) holds for j $\geq$ 1, $\sum_{k=1}^{j}(k2^k) = (j-1)2^{j+1} + 2$

RHS: $((j+1)-1)2^{(j+1)+1} + 2$

$(j)2^{j+2} + 2$

LHS: $\sum_{k=1}^{j}(k2^k) + (j+1)2^{j+1}$

$(j-1)2^{j+1} + 2 + (j+1)2^{j+1}$

$(2j)2^{j+1} + 2$

$(j)2^{j+2} + 2$

LHS = RHS therefore $p(k) \to p(k+1)$ holds for all n $\geq$ 1 by induction.

1.4 Question 18 - Prove by mathematical induction: Let P(n) be the statement that $n! < n^n$, where n is an integer greater than 1

P(n) = $n! < n^n$ for all n > 1

a) Basis step: P(2) = $2! < 2^2$

b) P(2) = $2! = 2 < 2^2 = 4$ which holds

Inductive step: $p(k) \rightarrow p(k+1)$

c) Assume p(k) holds for k > 1, $k! < k^k$

d) Need to prove that p(k) implies p(k+1) is true, meaning the LHS ¡ RHS for p(k+1)

e) RHS: $(k+1)^{k+1}$

$(k+1)(k+1)^k$

LHS:$(k+1)!$

$(k+1)(k)!$

$(k+1)(k)! < (k+1)(k^k)$ Application of the hypothesis

$(k+1)(k)! < (k+1)(k^k)$

Comparing LHS and RHS:

$(k+1)(k)! < (k+1)(k)^k < (k+1)(k+1)^k$ because k < k+1

f) These steps show under mathematical induction that starting with the smallest possible integer value for n greater than 1, (that is 2), the inequality holds. For any integer greater than or equal to 2 the inequality will continue to hold, as one integer implies the next holds the inequality.